SET NO. 23

EMPLOYEEPERFORMANCEANALYSIS

Name: SIVA KRISHNA.S

Registration No.: 1924260046

Roll No.: 37

Course:DSA0602 Data Handling and Visualization

Dataset: Employee Performance

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78
4	Sales	4	90
5	HR	2	76

Question 1: Department Distribution Bar Chart

Aim:

To visualize the number of employees in each department using a bar chart.

Algorithm:

1. Enter employee department data.

2. Count employees in each department.

3. Draw a bar chart.

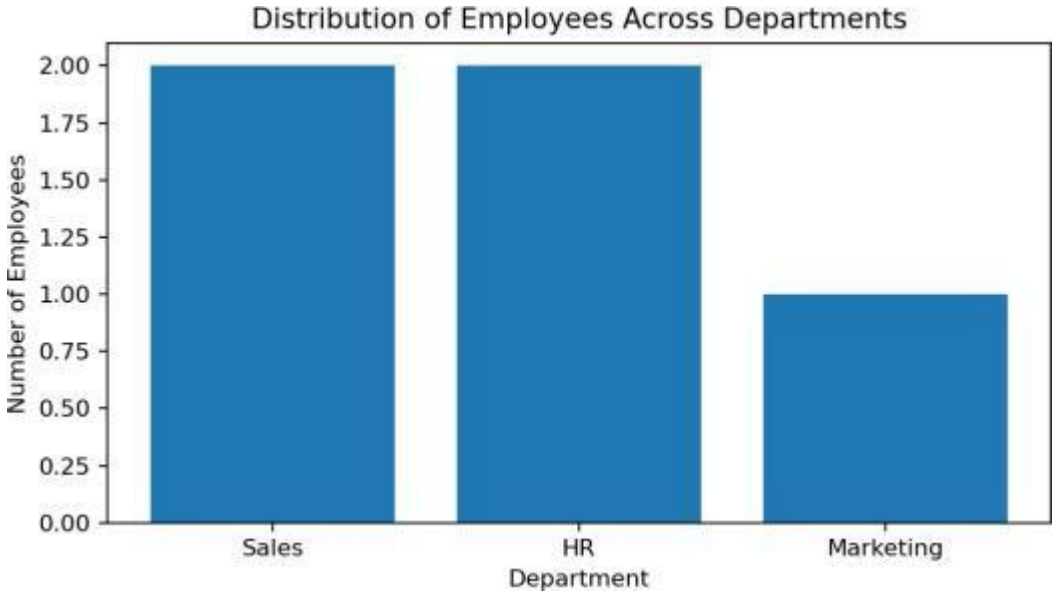
4. Label the x-axis, y-axis, and chart title.

Code (R):

```
department <- c("Sales", "HR", "Marketing", "Sales", "HR")
count <- table(department)

barplot(count, main="Distribution of Employees Across
    Departments", xlab="Department", ylab="Number of
    Employees",
col="skyblue")
```

Output: The bar chart shows Sales = 2 employees, HR = 2 employees, and Marketing = 1 employee.



Result: Sales and HR have the highest employee counts, while Marketing has one employee.

Question 2: Employee Performance Line Chart

Aim:

To display employee performance scores in a line chart.

Algorithm:

1. Store employee IDs and performance scores.

2. Plot the scores as a line.

3. Label both axes.

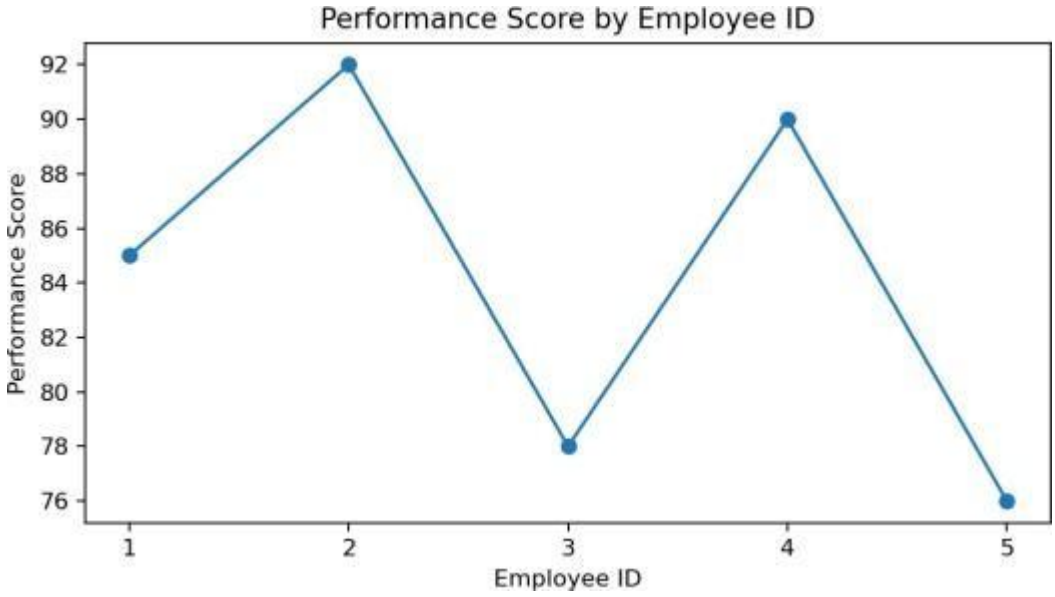
4. Add a title and display the chart.

Code (R):

```
employee_id <- c(1, 2, 3, 4, 5)
score <- c(85, 92, 78, 90, 76)

plot(employee_id, score, type="o", pch=19,
    main="Performance Score by Employee ID",
    xlab="Employee ID", ylab="Performance
    Score")
```

Output: A line chart displaying performance scores for employee IDs 1 to 5.



Result: Employee 2 has the highest score (92), and Employee 5 has the lowest score (76). A true performance trend over time requires dated observations.

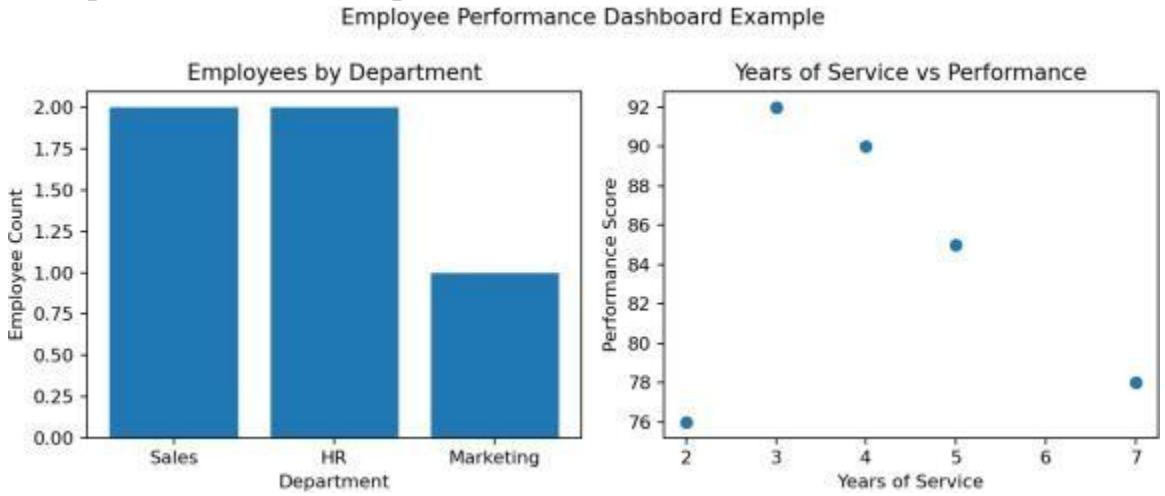
Question3: Tableau Dashboard

Aim: To build a dashboard combining department distribution and the relationship between years of service and performance score.

Algorithm:

1. Connect the employee dataset in Tableau.
2. Create a bar chart with Department and employee count.
3. Create a scatter plot with Years of Service on Columns and Performance Score on Rows.
4. Add Employee ID to Detail.
5. Place both worksheets on a dashboard and add a title.

Output: A dashboard example with a department bar chart and a years-of-service versus performance scatter plot.



Result: The dashboard compares department sizes and helps inspect whether years of service are associated with performance scores.

Question 4: Employee Performance Data Table

Aim: To display employee performance data in a labelled table using R.

Algorithm:

1. Create vectors for employee ID, department, years of service, and performance score.

2. Combine them into a data frame.

3. Display the data frame with column names.

Code (R):

```
employee_id <- c(1, 2, 3, 4, 5)
department <- c("Sales", "HR", "Marketing", "Sales", "HR")
years <- c(5, 3, 7, 4, 2) score <- c(85, 92, 78, 90, 76)
```

```
employee_data <-
data.frame( Employee_ID=
employee_id, Department = department,
Years_of_Service = years,
Performance_Score = score
)
print(employee_data)
```

Output: A table containing employee ID, department, years of service, and performance score.

Employee Performance Data Table

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78
4	Sales	4	90
5	HR	2	76

Result: The data frame presents all employee performance details in a clear, labelled table.